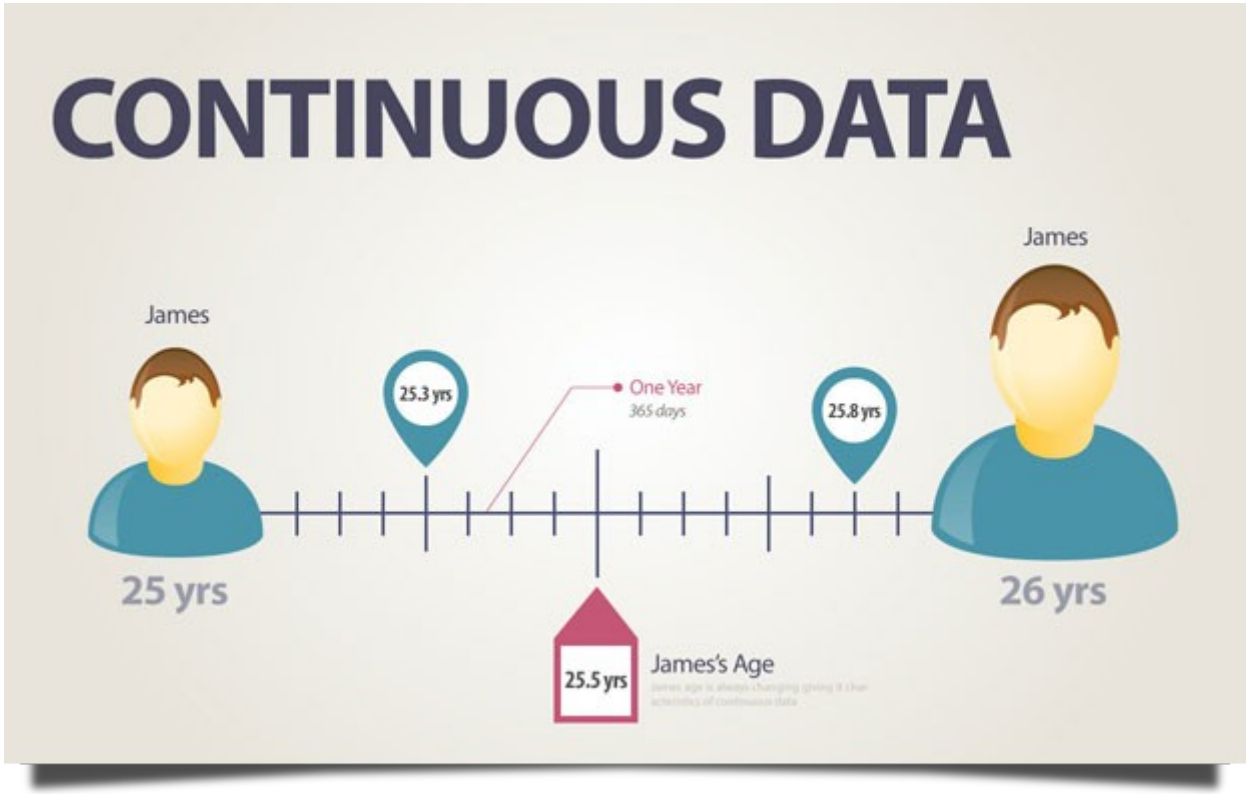
Lecture 2: Problems in Machine Learning

Is your data???

- Continuous or discrete?



How many days this week did you read the newspaper?

Number of days read	Frequency
0	44
1	24
2	18
3	16
4	20
5	22
6	26
7	30
Total	**200**

Is your data???

- Continuous or discrete?

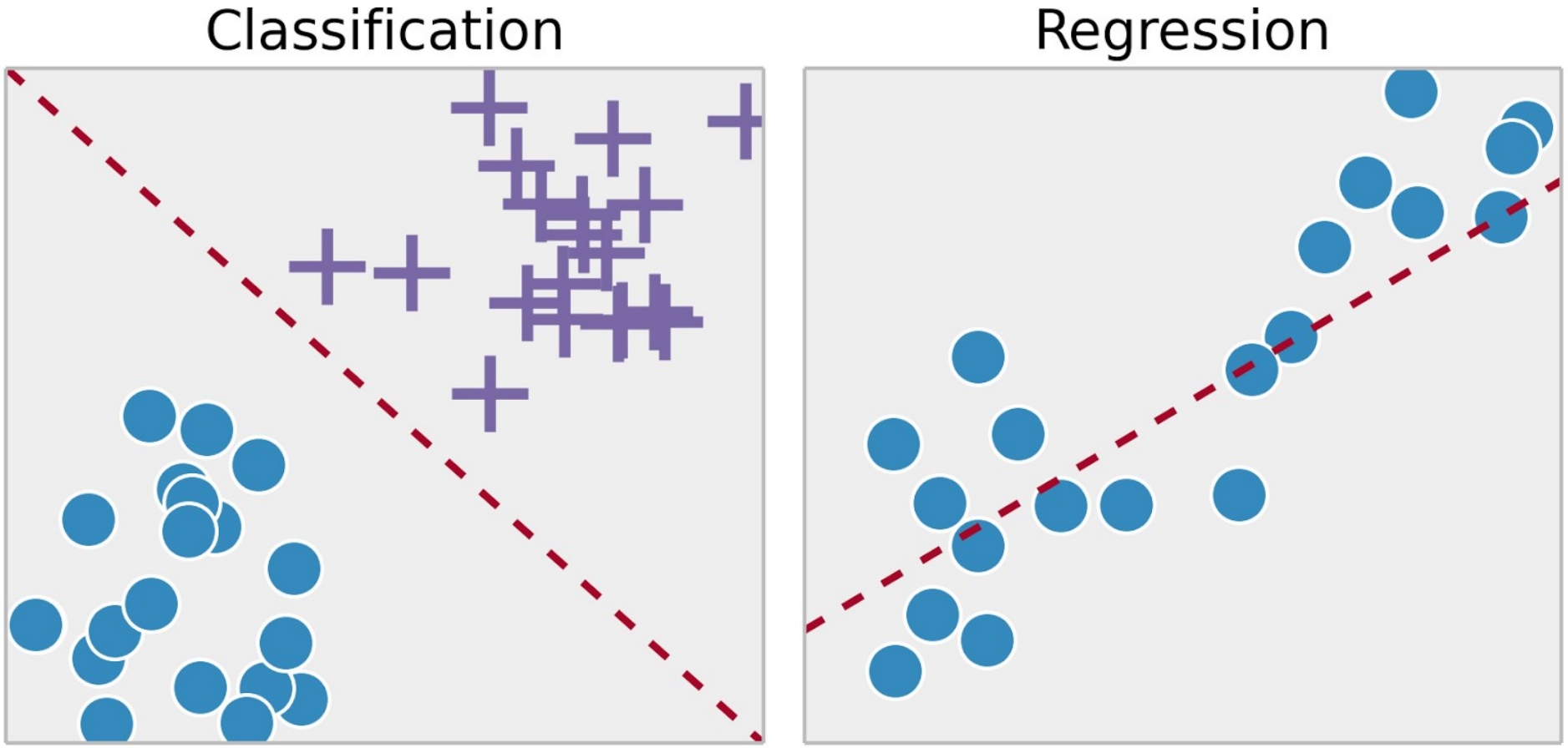
Differences between Continuous and Discrete Data	
Continuous	**Discrete**
Average Daily temperature	Days with temperature > 45 degree
No of minutes to board the plane	# of late departures due to late boarding
Number of litters of milk in the tank	Whether the tank is filled or not filled
Distance between the mile stones on a road	No of mile stones on a road

Is your data???

- Continuous or discrete – could you decide?

Is your data???

- Based on classification?



Is your data???

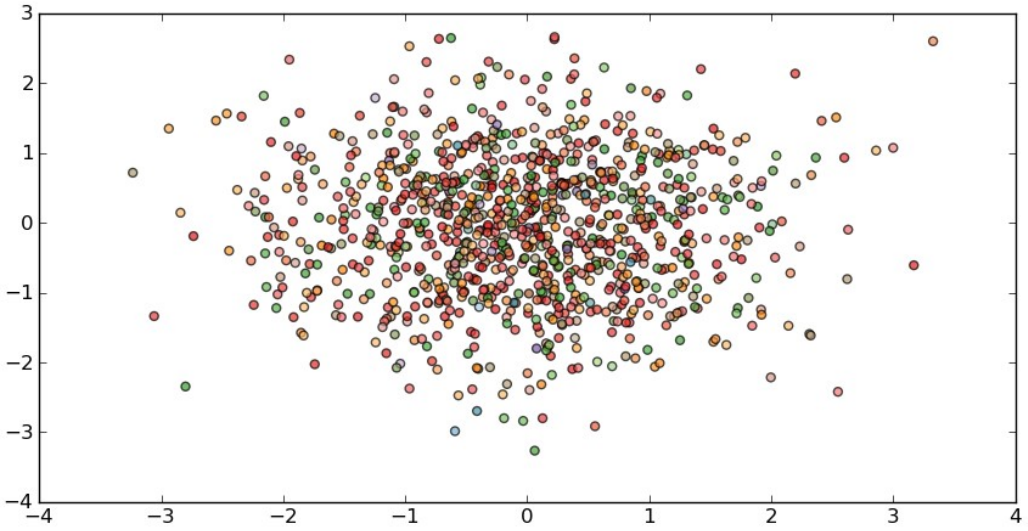
- Based on classification?
  - Spam vs not spam
  - Insure vs don’t insure

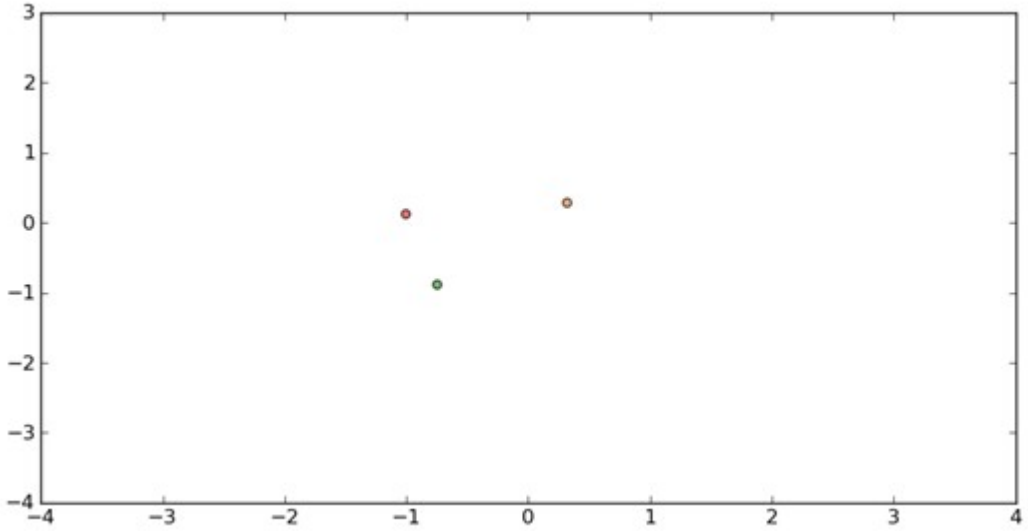
vs

  - Spam vs important vs “general”
  - Don’t insure, low rate, average rate, high rate

Is your data???

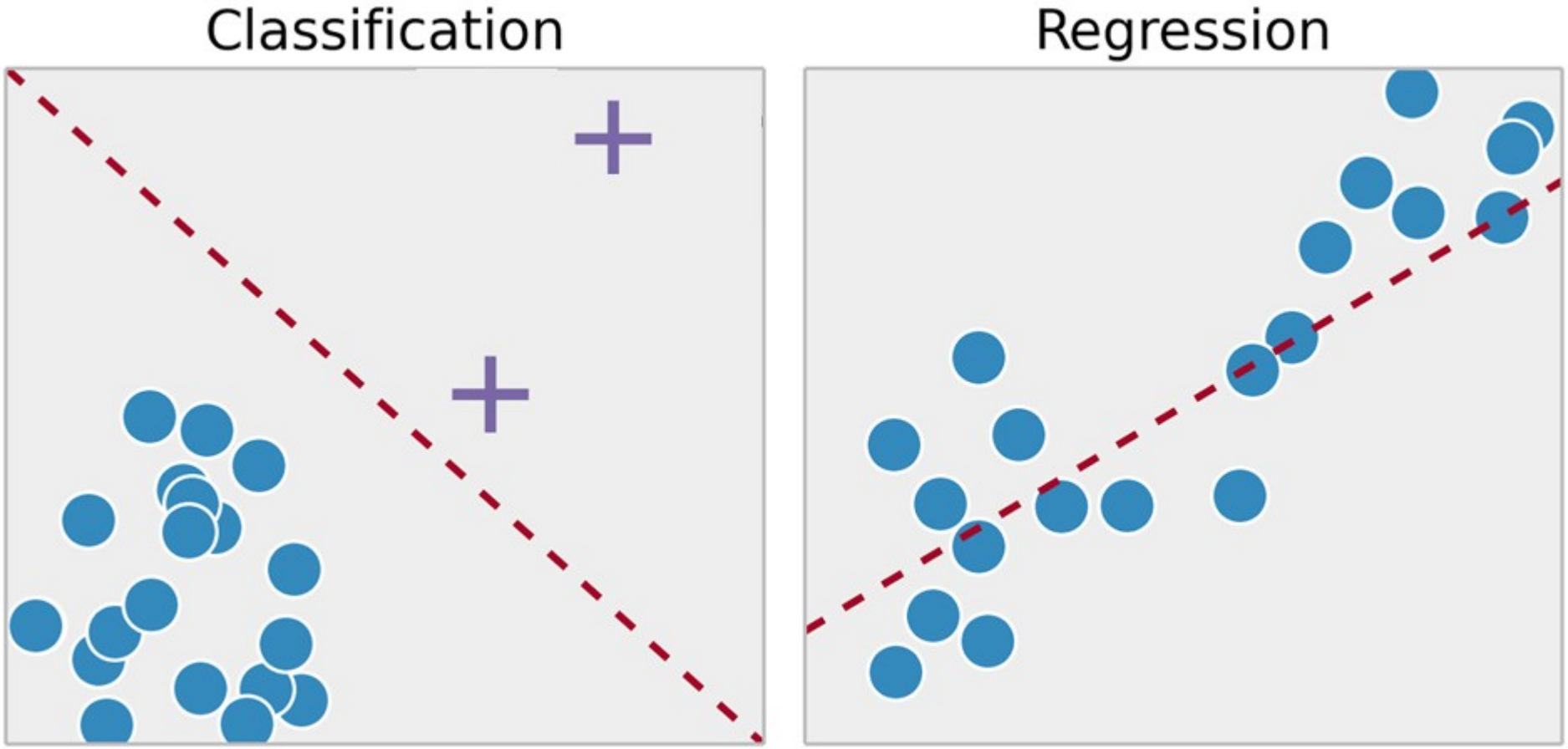
- Abundant vs rare





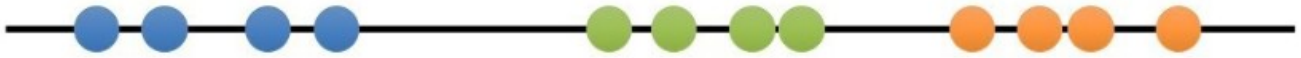
Is your data???

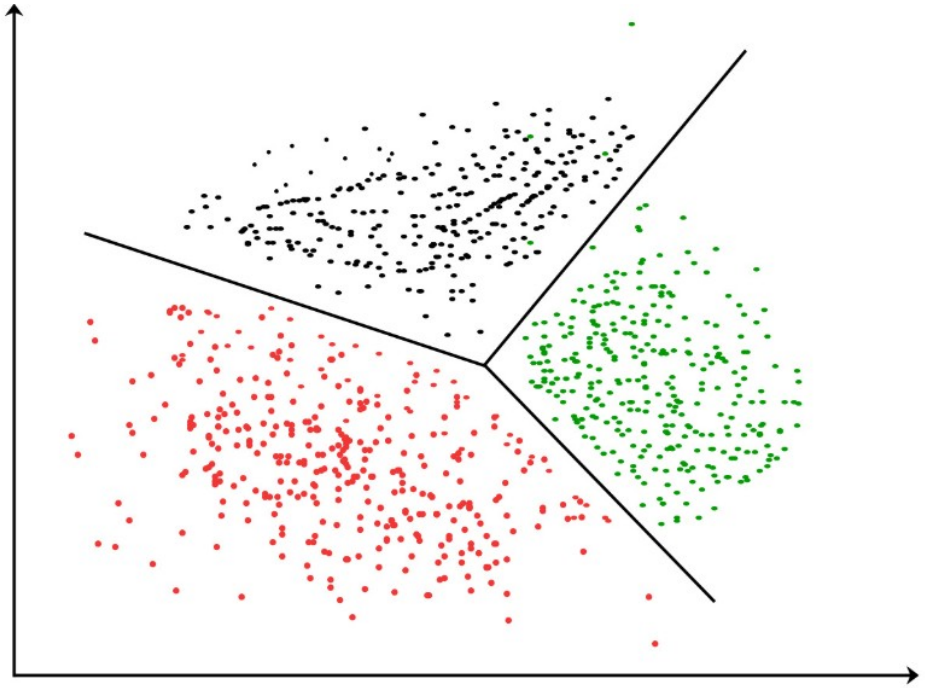
- Abundant vs rare – or does it depend on which kind of data?

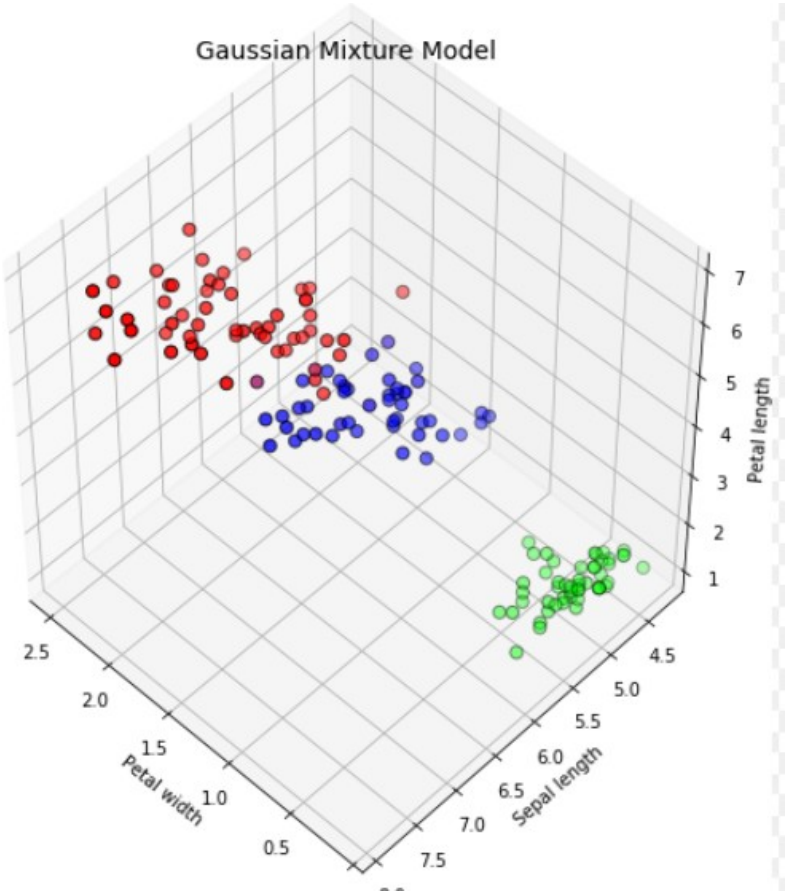


Is your data???

- Single- or multi-variable

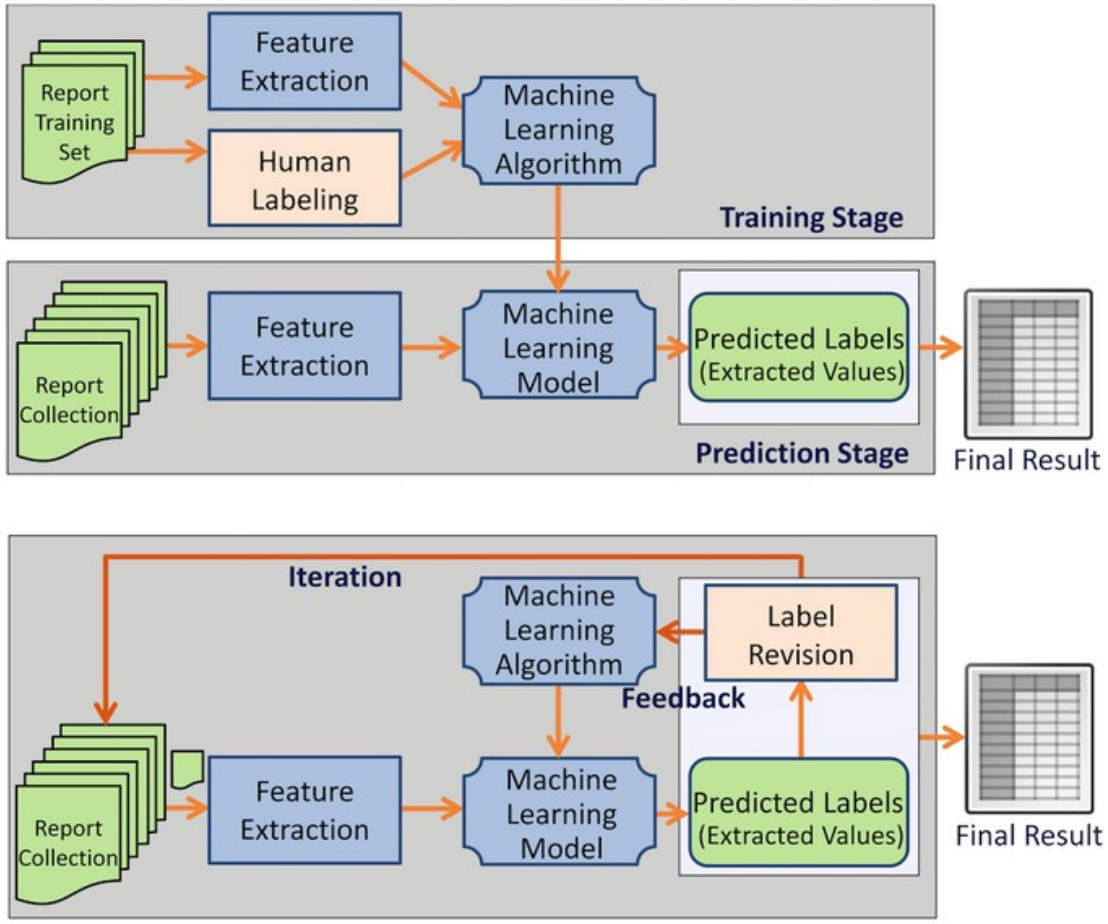






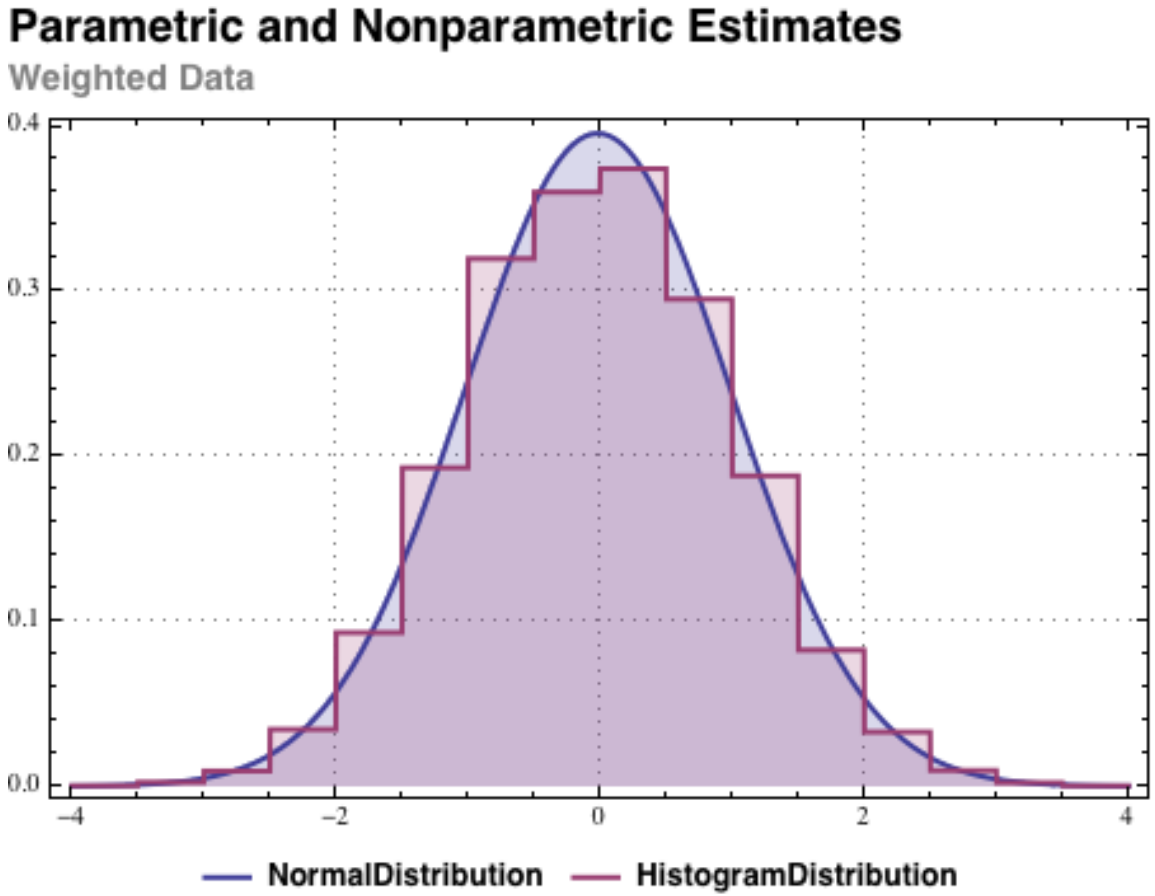
Is your data???

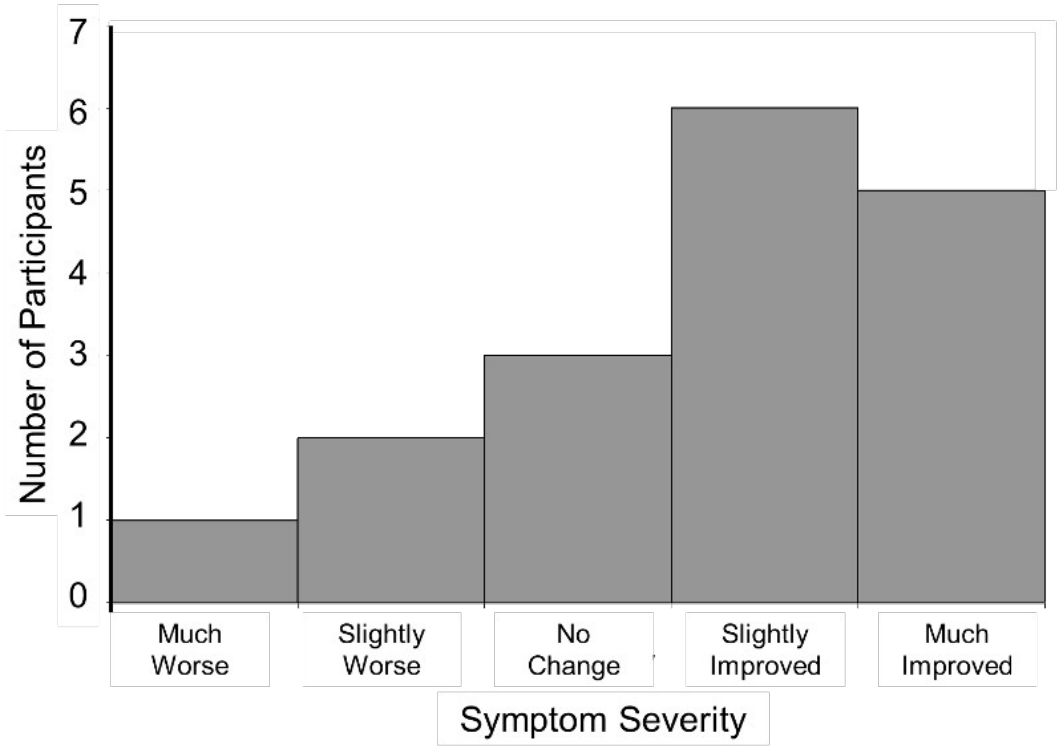
- Available all at once or given piece-by-piece



Is your data???

- Based upon an underlying statistical model?





Is your data???

- Based upon an underlying statistical model?

Is your data???

- You can (and will have to) combine all of these things (and more) just to *describe* your data

- Your choice of machine learning tool/algorithm will depend on all of these factors

- This diversity of data drives the diversity of solutions that exist to make sense of data